Requirements Document

Python-Based Research Intelligence and Benchmarking Assistant

1. Project Overview

I want to build a Python-based **Research Intelligence and Benchmarking Assistant**.

The purpose of the system is to automatically generate professional reports from a user-provided topic. These reports may include:

- State-of-the-art (SOA) analysis
- Market intelligence
- Competitor analysis
- Technology watch
- Patent landscape analysis
- Innovation gap analysis
- Benchmarking and opportunity analysis

The system should help R&D, product, innovation, patent/IP, and technical leadership teams quickly understand the existing landscape around a specific topic.

Example topics:

- AI-Based Debriefing in VR Medical Training
- Smart Surgical Mentor Using Generative AI
- Ultrasound Training in Virtual Reality
- AI-Driven Nursing Assessment
- Haptic Feedback for Medical Simulation

The user will provide a research topic, and the system should automatically search, collect, analyze, organize, summarize, and archive relevant information from multiple sources.

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2. Target Users

The final report should be suitable for:

- R&D teams
- Product teams
- Innovation teams
- Patent and IP teams
- Technical leadership
- Strategic decision-making teams
- Clinical stakeholders when relevant

- Educational stakeholders when relevant
- Business stakeholders when relevant

The system should support multiple stakeholder perspectives, including:

- R&D perspective
- Product perspective
- Clinical or educational perspective
- Business and market perspective
- Patent and IP perspective

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3. Information to Search and Collect

The system should search for and collect relevant information from the following categories:

1. Scientific papers
2. Patents
3. Commercial products and solutions from competitors relevant to the topic, including competitors of VirtualiSurg when applicable
4. Integratable tools, APIs, SDKs, libraries, frameworks, and platforms
5. Standards, guidelines, regulatory references, and industry best practices when relevant

The system should collect source links, metadata, summaries, images, figures, screenshots, PDFs, and related documents when available.

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4. Search and Ranking Requirements

The system should not only collect information but also rank and filter it.

The system should:

- Rank results by relevance score
- Remove duplicate results
- Prioritize peer-reviewed scientific papers over lower-quality sources
- Prioritize active patents over expired or abandoned patents
- Prioritize commercial solutions directly related to the topic
- Prioritize tools and APIs that can realistically be integrated into a software product
- Provide the reason why each result was selected
- Store the search queries used for traceability
- Store the source from which each result was collected

For each collected item, the system should include:

- Title or name
- Type of source

- Source URL
- Publication date or access date
- Author, inventor, assignee, company, or provider
- Short relevance explanation
- Confidence level

Confidence level should be one of:

- High
- Medium
- Low

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5. User Validation Step

Before generating the final report, the system should present an intermediate review step to the user.

The system should show:

- Proposed benchmark metrics
- List of collected scientific papers
- List of collected patents
- List of commercial products and solutions
- List of integratable tools, APIs, SDKs, libraries, frameworks, and platforms
- List of standards, guidelines, and regulatory references when applicable

The user should be able to:

- Approve the proposed benchmark metrics
- Edit the proposed benchmark metrics
- Remove irrelevant results
- Add missing results manually
- Rename report sections
- Confirm which items should be included in the final report

The system should only generate the final report after user approval.

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6. Evidence and Traceability Requirements

Every important statement in the report should be traceable to a source.

For each paper, patent, commercial solution, tool, API, library, or standard, the system should provide:

- Source URL
- Access date
- Evidence excerpt or short source note

- Confidence level
- Reference number

The system should avoid unsupported claims.

If information is missing or unclear, the system should explicitly write:

- "No public information found"
- "Not mentioned in available sources"
- "Information could not be verified from public sources"

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7. Report Formatting Requirements

The generated report should follow the style and structure of the sample report provided by the user.

General formatting requirements:

- Font: Calibri
- Text alignment: Justify or Full Justify
- Main theme color: `#134f5c`
- All report text should be written clearly and professionally
- Language should be simple, direct, and easy to understand
- Tables should be clean, readable, and consistent
- The report should use numbered headings

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8. Report Structure

The report should include the following sections.

8.1 Cover Page

The cover page should include:

- Company logo

- Center aligned

- Report title

- Center aligned
- Font size: 16
- Bold

- Color code: `#134f5c`

- Section: “What is this document about?”

- Font size: 11
- Bold

- Color: black

- Short explanation of the report purpose

- Font size: 11

- Color: black

- Change Tracking table

The Change Tracking table should use the following format:

Version	Date	Change	Modified By
1.X	[date]	Creation of the document	[take the name from input]

Table formatting:

- Header background color: `#134f5c`
- Header font color: white
- Header font: bold
- Table body font color: black

The cover page should also include basic document metadata:

- Author
- Generated date
- Topic
- Version
- Number of papers analyzed
- Number of patents analyzed
- Number of commercial solutions analyzed
- Number of tools analyzed

8.2 Table of Contents

The report should include an automatically generated Table of Contents.

Formatting:

- Heading font size: 14
- Bold
- Color code: `#134f5c`

The Table of Contents should include links to all major sections and subsections.

8.3 Benchmarking and Gap Analysis

Formatting:

- Heading level: Heading 1
- Font color: `#134f5c`
- Bold
- Numbered section

The system should automatically identify meaningful comparison metrics based on the topic.

Before generating the final report, the system should suggest these metrics to the user and ask for verification.

The benchmarking table should compare all identified papers, patents, commercial solutions, and tools.

The table should follow this general format:

Company / Case	Metric 1	Metric 2	Metric 3	Metric 4	Metric 5	Metric 6	Metric 7	%
Case 1	✔️	✔️	❌	✔️	❌	✔️	✔️	0.60
Case 2	❌	✔️	❌	❌	✔️	✔️	✔️	0.60
Case 3	✔️	✔️	❌	❌	✔️	✔️	✔️	0.60
Case n	❌	❌	❌	❌	❌	❌	❌	0.00
% Features	0.22	0.72	0.22	0.05	0.30	0.38	0.66	—

Legend:

- ✔️ Fully Matched / Covered
- ❌ Not Covered
- ⚠️ Partially Covered

Coverage values should be mapped as:

- Covered = 1.0
- Partial = 0.5
- Not Covered = 0.0

Final score should be calculated as:

Final Score = Sum of metric scores / Number of metrics

The results should be sorted by final score percentage, from highest to lowest.

Background color should reflect score percentage:

- Higher score: darker green
- Medium score: lighter green or yellow
- Lower score: orange or light red
- Zero or very low score: red or light red

This section should include:

- Opportunity analysis
- Innovation gaps
- Competitive differentiation opportunities
- Unmet needs
- White spaces
- Potential innovation directions

For every opportunity, the system should provide:

- Opportunity title
- Description
- Expected value
- Estimated implementation complexity: Low, Medium, or High

8.4 Executive Summary

Formatting:

- Heading level: Heading 1
- Font color: `#134f5c`
- Bold
- Numbered section

For every identified paper, patent, commercial solution, and tool, generate a concise executive summary.

Each summary item should follow this format:

“[Name] integrates/proposes/provides [main capabilities] to support [main purpose] (read more).”

Requirements:

- Each summary should be maximum one and a half lines
- The “read more” text should link to the detailed section of that item
- Summaries should be grouped by category:
- Commercial Products and Solutions
- Scientific Papers
- Patents
- Integratable Tools
- Standards and Guidelines when applicable
- Summaries should be sorted by relevance score
- The Top 5 most relevant findings should be highlighted

This section should also include:

- Key findings
- Strategic insights
- Emerging trends
- Recommended opportunities for innovation and differentiation

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9. State-of-the-Art Section

Formatting:

- Heading level: Heading 1
- Font color: `#134f5c`
- Bold
- Numbered section

The State-of-the-Art section should include detailed analysis sections for:

1. Commercial Products and Solutions
2. Scientific Papers
3. Patents
4. Integratable Tools, APIs, SDKs, Libraries, and Platforms
5. Standards, Guidelines, and Regulatory Considerations when applicable

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9.1 Commercial Products and Solutions

Formatting:

- Heading level: Heading 2
- Font size: 13
- Bold
- Color code: `#134f5c`

For each commercial product or solution:

- Use the company name as the section title
- Provide an overview of the company and relevant product
- Explain why the product is related to the topic
- Include related images when available
- Include short explanatory captions for images
- Present product features in a simple and reader-friendly table
- Include strengths and limitations
- Include competitive positioning when applicable
- Include a summary table using the same metrics as the benchmarking section

Subsection formatting:

- Heading level: Heading 3
- Font size: 12
- Bold
- Color code: `#134f5c`

Example structure:

9.1.1 Case / Company Name

Overview

Explain the company, product, and relevance to the topic.

Features

Category	Feature Name	Description
[Category]	[Feature]	[Description]

Images and Figures

When images are available:

- Download and archive images

- Add figure numbers
- Add captions
- Add source references

Example:

Figure X. UbiSim learner performance dashboard.
Source: UbiSim website.

Summary

Metric	Details	Coverage
Metric A	Description	Covered
Metric B	Description	Partial
Metric C	Description	Not Covered

Coverage values must be limited to:

- Covered
- Partial
- Not Covered

9.2 Scientific Papers

Formatting:

- Heading level: Heading 2
- Font size: 13
- Bold
- Color code: `#134f5c`

For each scientific paper:

- Use the paper title as the section title
- Explain why the paper is related to the topic
- Explain the paper’s contribution
- Explain the method in simple words
- List the relevant results in a very simple way
- Include figures or diagrams when useful and available
- Include a summary table using the same benchmark metrics

Subsection formatting:

- Heading level: Heading 3

- Font size: 12
- Bold
- Color code: `#134f5c`

Example structure:

9.2.1 Paper Title

Context

Explain the paper contribution and why the paper is related to the topic.

Method

Explain how the authors implemented or tackled the problem.

Step	Description
Data Collection	Description
Model / System Design	Description
Evaluation	Description

Results

List the related results in a very simple way.

Summary

Metric	Details	Coverage
Metric A	Description	Covered
Metric B	Description	Partial
Metric C	Description	Not Covered

Coverage values must be limited to:

- Covered
- Partial
- Not Covered

9.3 Patents

Formatting:

- Heading level: Heading 2
- Font size: 13
- Bold
- Color code: `#134f5c`

For each patent:

- Use the assignee name as the section title
- Provide patent title
- Provide patent number
- Provide publication date
- Provide grant date when available
- Provide status
- Provide expiration date when available
- Explain the invention in simple words
- Include an image from the patent whenever possible
- Explain the selected figure
- Summarize the claims in simple words
- Include a summary table using the same benchmark metrics

Subsection formatting:

- Heading level: Heading 3
- Font size: 12
- Bold
- Color code: `#134f5c`

Example sentence:

“[Assignee Name] published a patent titled ‘[Patent Title]’ in [Date], which describes [brief description of the invention].”

Patent Lifecycle Overview

Tech. Type	Country	Grant	Status	Expiration
[tech-type]	[Country name]	[Date]	[Granted, Abandoned, Pending, or other]	[Date]

Claims

Claim	Description
1	Description

Claim	Description
2	Description

Summary

Metric	Details	Coverage
Metric A	Description	Covered
Metric B	Description	Partial
Metric C	Description	Not Covered

Coverage values must be limited to:

- Covered
- Partial
- Not Covered

9.4 Integratable Tools, APIs, SDKs, Libraries, and Platforms

Formatting:

- Heading level: Heading 2
- Font size: 13
- Bold
- Color code: `#134f5c`

For each tool, API, SDK, library, framework, or platform:

- Name
- Type
- Provider
- Source link
- Description
- Integration capabilities
- Key features
- Relevance to the topic
- Strengths
- Limitations
- License or pricing note when available
- Technical maturity when available
- Summary table using the same benchmark metrics

Example structure:

Tool / Platform Name

Overview

Explain what the tool is and why it is relevant.

Integration Capabilities

Explain how it can be integrated into a software product.

Features

Feature	Description
Feature A	Description
Feature B	Description

Summary

Metric	Details	Coverage
Metric A	Description	Covered
Metric B	Description	Partial
Metric C	Description	Not Covered

Coverage values must be limited to:

- Covered
- Partial
- Not Covered

9.5 Standards, Guidelines, and Regulatory Considerations

This section should be included when applicable.

Formatting:

- Heading level: Heading 2
- Font size: 13
- Bold
- Color code: `#134f5c`

This section should include:

- Relevant standards
- Guidelines
- Regulatory references
- Industry best practices
- Compliance considerations
- Relevance to the topic

Examples may include:

- Healthcare standards
- Data privacy regulations
- AI governance frameworks
- Clinical education standards
- Simulation training standards

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10. References

Formatting:

- Heading level: Heading 1
- Font color: `#134f5c`
- Bold
- Numbered section

The system should include all sources used in IEEE format.

References should include:

- Papers
- Patents
- Company websites
- Product pages
- Technical documentation
- APIs
- Libraries
- Standards
- Guidelines
- Regulatory references

Each reference should include:

- Reference number
- Title
- Author, company, inventor, or organization

- Year or publication date when available
- URL
- Access date

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11. Google Drive Integration

The system should automatically create a structured Google Drive workspace for each research topic.

The main folder name should be generated automatically based on the topic and date.

Example:

Research - AI-Based Debriefing in VR Medical Training - YYYY-MM-DD

Inside the main folder, the system should create the following subfolders:

- Scientific Papers
- Patents
- Commercial Solutions
- Integratable Tools
- Standards and Regulations
- Final Report

11.1 Scientific Papers Folder

The system should:

- Download and store all available paper PDFs
- Rename each document with the title of the paper
- Store metadata files
- Store figures
- Store screenshots
- Store notes

11.2 Patents Folder

The system should:

- Download and store all available patent PDFs
- Rename each patent with the assignee name and patent title
- Store patent figures

- Store claim summaries
- Store patent metadata
- Store notes

11.3 Commercial Solutions Folder

The system should create a dedicated folder for each company.

Each company folder should store:

- Screenshots
- Product brochures
- Images
- Notes
- Supporting files
- Web captures when available

Example:

```
Commercial Solutions/
├── UbiSim/
├── Osso VR/
├── SimX/
└── Lumeto/
```

11.4 Integratable Tools Folder

The system should create a dedicated folder for each tool, API, SDK, library, framework, or platform.

Each folder should store:

- Documentation
- Images
- Example files
- Notes
- Supporting files

Example:

```
Integratable Tools/
├── LangChain/
├── Semantic Scholar API/
├── Google Patents/
└── OpenAlex/
```

11.5 Standards and Regulations Folder

The system should store:

- Standards documents and references
- Regulatory guidance references
- Compliance notes
- Supporting documentation
- Related images or files

11.6 Final Report Folder

The system should store:

- Markdown report
- DOCX report
- PDF report
- CSV benchmark table
- Excel benchmark table
- References file
- Supporting report assets

11.7 Index File

The system should create an `index.md` file inside the main Google Drive folder.

The `index.md` file should include:

- Research topic
- Creation date
- Last update date
- Folder structure
- Collected papers
- Collected patents
- Commercial solutions
- Tools and platforms
- Standards and regulatory references
- Links to all archived materials
- Link to the final report

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12. Output Formats

The system should generate:

- Markdown report
- DOCX report
- PDF report
- CSV benchmark table
- Excel benchmark table
- Google Drive archive
- `index.md` file

The first version can generate Markdown first.

Later versions should support:

- DOCX export
- PDF export
- Google Docs export
- Excel export

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13. Research Workspace History

The system should maintain a history of generated research workspaces.

For each topic, the system should store:

- Research topic
- Creation date
- Last update date
- Report version
- Number of papers analyzed
- Number of patents analyzed
- Number of commercial solutions analyzed
- Number of tools analyzed
- Google Drive folder link
- Final report link

The user should be able to regenerate or update an existing report.

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14. Additional Requirements

The generated report should contain hyperlinks to all supplementary materials stored in Google Drive, including:

- Papers
- Patents
- Images
- Screenshots
- Commercial solution folders
- Tool documentation
- Standards and regulatory references
- Supporting documents

The system should ask for user confirmation before:

- Creating the final report
- Downloading large files
- Creating Google Drive folders
- Exporting PDF or DOCX files
- Updating an existing report

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15. Development Workflow

The system should be built using spec-based development.

Before writing any code, the AI development tool should:

1. Create `requirements.md`
2. Ask me to review and approve it
3. Create `spec.md`
4. Ask me to review and approve it
5. Create `tasks.md`
6. Ask me to review and approve it
7. Only then start implementation

Do not generate code before the requirements, specification, and tasks are reviewed and approved.

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16. Assumptions

The system may assume:

- The user has permission to use the company logo
- The user has access to Google Drive
- Some sources may not allow automatic PDF downloads
- Some commercial websites may block scraping
- Some patent or paper databases may require API keys
- Some standards or regulatory documents may not be freely downloadable
- Some generated reports may require manual review before business use

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17. Risks

The system should consider the following risks:

- Search results may be incomplete
- Some sources may contain outdated information
- Some commercial claims may be marketing-based and not independently verified
- Some patents may be difficult to classify automatically
- Some PDFs may not be downloadable
- Some APIs may have rate limits
- Some sources may require authentication
- AI-generated summaries may contain errors if not grounded in sources
- Google Drive API permissions may require configuration

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18. Dependencies

The system may depend on:

- Google Drive API
- Google Docs API
- Google Sheets API
- Scientific paper APIs such as Semantic Scholar, OpenAlex, Crossref, PubMed, or arXiv
- Patent data sources such as Google Patents, PatentsView, Lens.org, or Espacenet
- Web search or scraping tools
- PDF extraction tools
- Screenshot or image capture tools
- Markdown, DOCX, PDF, CSV, and Excel generation libraries
- LLM API for summarization and analysis

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19. Open Questions

Before creating the technical specification, the following questions should be clarified:

1. Should the first version support only Markdown, or also DOCX and PDF?
2. Which paper APIs should be used first?
3. Which patent source should be used first?
4. Should Google Drive integration be included in version 1?
5. Should the system create Google Docs directly, or only upload generated files to Google Drive?
6. Should the company logo be provided manually by the user?
7. Should the benchmark metrics be fully AI-generated or selected from predefined metric templates?
8. Should the system support manual editing of results before final report generation?
9. Should the system support multiple languages?
10. Should the system run as a command-line tool, web app, or desktop app?
11. Should the system store research history locally, in Google Drive, or in a database?
12. Should the system automatically download PDFs, or only store links when download is restricted?
13. Should the system include screenshots from commercial websites?
14. Should the system support citation verification before final export?
15. Should the system include a confidence score for every extracted claim?

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20. First Version Scope

For the first version, the system should focus on:

- User enters a topic
- System searches for papers, patents, commercial solutions, and tools
- System proposes benchmark metrics
- User reviews and approves the metrics and selected sources
- System generates a Markdown report
- System creates a basic Google Drive folder structure
- System stores collected links and available documents
- System creates an `index.md` file

PDF, DOCX, Google Docs, Excel export, and advanced automation can be added in later versions.